

DAVAO FRANCISCO BANGOY INTL, Philippines

WMO: 987530

Lat: 7.126N Lon: 125.646E Elev: 29 StdP: 100.97 Time Zone: 8.00 (E08) Period: 94-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	23.2	20.0	14.7	27.1	20.8	15.5	26.7	8.5	26.0	6.8	29.4	1.9	0	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.1	34.1	26.8	33.5	26.7	33.1	26.7	28.2	32.3	27.8	31.9	27.6	31.6	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	22.9	31.3	26.8	22.5	31.0	26.2	21.8	30.3	91.2	32.3	89.5	32.0	88.0	32.0	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.7	5.8	5.0	DB	21.1	35.7	1.5	1.2	20.1	36.5	19.2	37.2	18.4	37.9	17.3	38.7	
			WB	19.7	29.3	1.6	0.6	18.6	29.7	17.6	30.1	16.7	30.4	15.5	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	27.3	27.6	28.2	28.8	28.8	28.3	28.1	28.2	28.3	28.5	28.2	28.0
	DBStd	1.02	1.23	1.06	1.06	0.93	0.89	0.85	0.81	0.81	0.76	0.70	0.90	1.03
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6641	537	492	563	566	583	549	561	565	548	573	547	557
	CDD18.3	3599	278	259	304	316	325	299	302	307	298	315	297	299
	CDH23.3	38499	2590	2514	3196	3636	3731	3292	3257	3372	3259	3443	3107	3103
(13)	CDH26.7	14638	845	877	1229	1536	1520	1256	1188	1273	1263	1387	1152	1112
(14)	Wind	WSAvg	2.3	2.6	2.9	2.7	2.4	2.1	1.9	2.0	2.2	2.0	2.1	2.4
(15) Precipitation	PrecAvg	1805	118	127	92	163	216	163	167	174	196	153	137	102
	PrecMax	2254	249	259	200	326	462	314	388	348	329	228	247	227
	PrecMin	1129	4	20	15	29	47	21	34	38	61	74	40	29
	PrecStd	293	65	79	58	93	107	86	93	83	84	42	61	51
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.3	34.2	35.0	34.4	33.9	33.2	33.3	33.8	34.1	33.8	33.5
		MCWB	26.3	26.0	25.9	26.3	27.1	27.5	26.9	26.8	27.1	27.3	27.0	26.8
	2%	DB	32.2	32.8	33.2	34.0	33.4	32.9	32.2	32.5	32.9	33.1	33.0	32.9
		MCWB	26.1	26.0	25.9	26.6	27.1	27.3	26.7	26.7	27.0	27.0	26.9	26.7
	5%	DB	31.8	32.0	32.8	33.2	32.9	32.1	31.9	32.0	32.1	32.5	32.2	32.1
		MCWB	26.0	25.8	25.9	26.7	27.1	26.9	26.6	26.6	26.7	26.8	26.6	26.5
	10%	DB	30.8	31.1	31.9	32.5	32.0	31.4	31.1	31.2	31.5	31.9	31.8	31.2
		MCWB	25.9	25.8	25.9	26.5	26.9	26.7	26.6	26.5	26.6	26.6	26.6	26.4
	0.4%	WB	27.6	27.7	27.9	28.4	28.6	28.4	28.0	28.1	28.1	28.4	28.2	28.1
		MCDB	31.6	31.8	32.5	32.9	32.7	32.3	31.8	31.8	32.1	32.8	32.3	32.2
	2%	WB	27.0	27.0	27.2	27.7	28.0	27.7	27.4	27.5	27.6	27.7	27.6	27.4
		MCDB	30.9	31.0	31.6	32.2	32.0	31.6	31.2	31.2	31.6	31.9	31.7	31.4
	5%	WB	26.5	26.5	26.7	27.3	27.6	27.3	27.1	27.1	27.2	27.3	27.2	27.0
		MCDB	30.2	30.4	30.9	31.6	31.4	31.0	30.7	30.7	31.0	31.4	31.1	30.8
	10%	WB	26.1	26.1	26.3	26.9	27.2	27.0	26.7	26.7	26.8	26.9	26.7	26.6
		MCDB	29.6	29.9	30.3	31.1	30.9	30.5	30.1	30.2	30.5	30.9	30.5	30.2
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	7.9	8.1	7.5	7.1	7.0	7.2	7.5	8.0	7.5	7.3
		MCDBR	8.3	8.7	9.1	8.9	8.3	7.9	7.7	7.8	8.1	8.7	8.4	8.4
		MCWBR	3.6	3.8	3.8	3.5	3.5	3.6	3.5	3.5	3.8	3.9	3.7	3.6
	5% WB	MCDBR	7.4	7.8	8.1	8.2	7.6	7.4	7.2	7.3	7.6	8.1	7.9	7.8
		MCWBR	3.5	3.7	3.8	3.6	3.6	3.6	3.6	3.6	3.8	4.0	3.8	3.6
(40) Clear-Sky Solar Irradiance	taub		0.407	0.404	0.403	0.412	0.410	0.399	0.406	0.424	0.426	0.446	0.409	0.406
	taud		2.498	2.489	2.500	2.468	2.491	2.531	2.501	2.431	2.413	2.358	2.490	2.501
	Ebn at Noon		909	920	919	896	880	879	876	873	884	869	900	903
	Edh at Noon		109	113	113	115	109	103	107	117	121	127	109	107
(44) All-Sky Solar Radiation	RadAvg		4.23	4.88	5.35	5.59	5.26	4.93	4.91	5.10	5.25	5.07	4.63	4.37
(45)	RadStd		0.43	0.47	0.45	0.38	0.27	0.22	0.27	0.27	0.28	0.31	0.39	0.33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.24	+0.27	N/A	+0.41	+0.45	+0.40	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.24	N/A	N/A	N/A	N/A